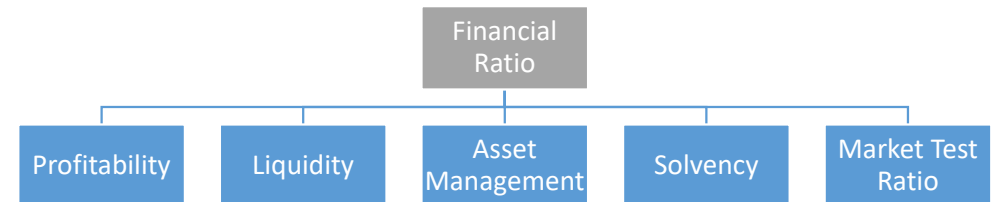


Financial Analysis

Financial Ratio Analysis

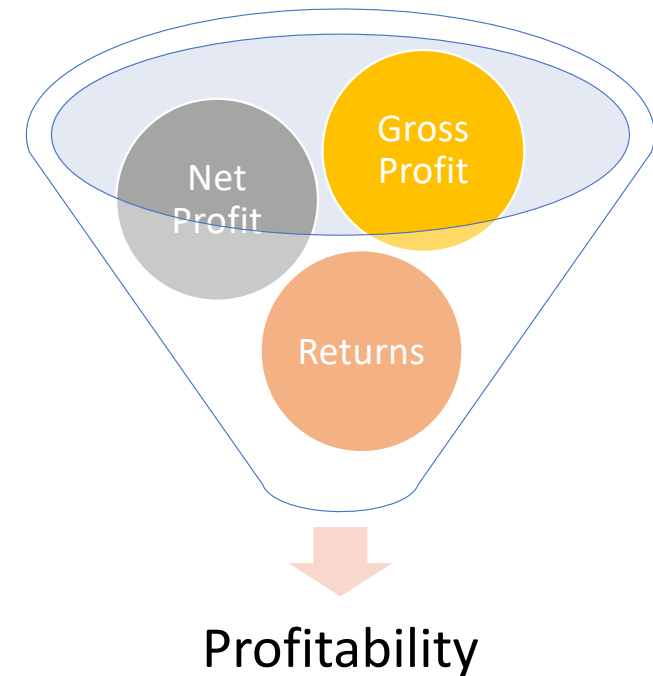
- Financial ratio analysis involves calculating and analyzing ratio that use data from financial statements.
- Ratio analysis express relationship between different financial statements.
- Financial ratio can be classified across five categories:
 - Profitability ratio
 - Liquidity ratio
 - Asset management or Activity ratio
 - Solvency ratio
 - Market test ratio



Profitability Ratio

Three elements of profitability analysis:

- Analyzing on sales and trade margins – Focus on Gross profit
- Analyzing on controlling expense – Focus on Net profits
- Analyzing the return on assets, return on equity and return on capital employed.



Gross Profit Ratio

- The gross profit ratio is an accurate indicator of how efficiently a company is selling its goods and services.
- This will give the organization's management and potential investors a view of how well the company manages to optimize its processes, keep the costs to a minimum and produce the highest possible profit.
- The formula for gross profit ratio is:

$$\text{Gross Profit} = \frac{\text{Revenue} - \text{COGS}}{\text{Revenue}} \times 100$$

Net Profit Ratio

- Net profit margin measures how much net income is generated as a percentage of revenues received.
- Net profit margin helps investors assess if a company's management is generating enough profit from its sales and whether operating costs and overhead costs are being contained.
- Net profit margin is one of the most important indicators of a company's overall financial health.

$$\text{Net Profit margin} = \frac{R - \text{COGS} - E - I - T}{\text{Revenue}} \times 100$$

Where:

R = Revenue

COGS = Cost of goods sold

E = Operating and other expenses

I = Interest

T = Taxes

Return on Equity

- Return on equity (ROE) measures a corporation's profitability in relation to stockholders' equity.
- Whether an ROE is considered satisfactory will depend on what is normal for the industry or company peers.
- As a shortcut, investors can consider an ROE near the long-term average of the Game Index (14%) as an acceptable ratio and anything less than 10% as poor.

$$\text{Return on Equity} = \frac{\text{Net Income}}{\text{Shareholder Equity}} \times 100$$

Return on Assets

- Return on assets (ROA) is an indicator of how well a company utilizes its assets in terms of profitability.
- ROA is best used when comparing similar companies or by comparing a company to its own previous performance.
- ROA does not take into account a company's debt, while return on equity (ROE) does—if a company carries no debt, its shareholders' equity and its total assets will be the same and ROA would equal ROE.

$$\text{Return on Assets} = \frac{\text{Net Income}}{\text{Total Assets}} \times 100$$

Return on Capital Employed

- ROCE is a metric for analysing profitability, and potentially comparing profitability levels across companies in terms of capital. There are two components required to calculate return on capital employed: earnings before interest and tax and capital employed.
- EBIT, also known as operating income, shows how much a company earns from its operations alone without regard to interest or taxes.
- Capital employed is found by subtracting current liabilities from total assets, which ultimately gives you shareholders' equity plus long-term debts.

$$\text{ROCE} = \frac{\text{EBIT}}{\text{Capital Employed}} \times 100$$

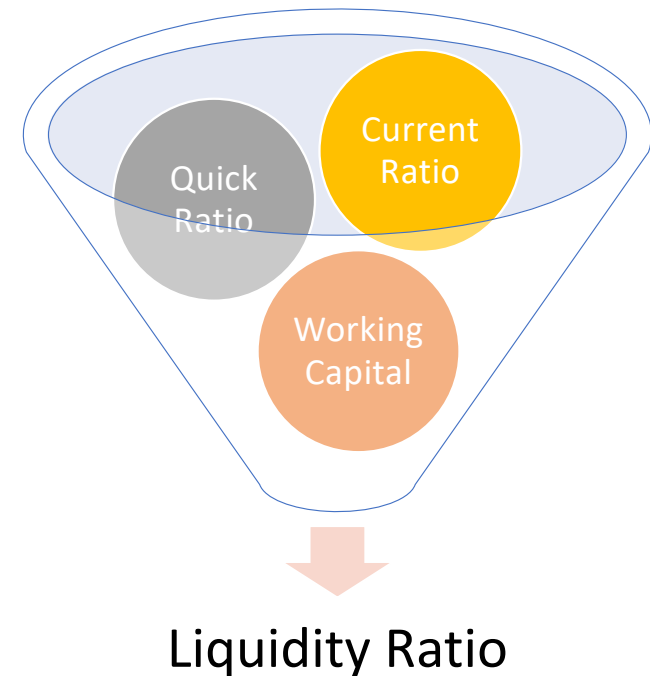
Where:

EBIT = Earnings before Interest & Tax

Capital Employed = Total assets - Current liability

Liquidity Ratio

- Liquidity ratios are class of financial metrics used to determine a debtor's ability to pay off current debt obligations without raising external capital.
- Common liquidity ratios include the quick ratio, current ratio, and working capital.
- Liquidity ratios determine a company's ability to cover short-term obligations and cash flows, while solvency ratios are concerned with a longer-term ability to pay ongoing debts.



Current Ratio

- The current ratio compares all of a company's current assets to its current liabilities.
- These are usually defined as assets that are cash or will be turned into cash in a year or less, and liabilities that will be paid in a year or less.
- The current ratio helps investors understand more about a company's ability to cover its short-term debt with its current assets and make apples-to-apples comparisons with its competitors and peers.
- Weaknesses of the current ratio include the difficulty of comparing the measure across industry groups, overgeneralization of the specific asset and liability balances

$$\text{Current Ratio} = \frac{\text{Current Assets}}{\text{Current Liability}}$$

Quick Ratio/Acid Test Ratio

- The quick ratio measures a company's capacity to pay its current liabilities without needing to sell its inventory or obtain additional financing.
- The quick ratio is considered a more conservative measure than the current ratio, which includes all current assets as coverage for current liabilities.
- The higher the ratio result, the better a company's liquidity and financial health; the lower the ratio, the more likely the company will struggle with paying debts.

$$\text{Cash Ratio} = \frac{\text{CE}}{\text{Current Liability}}$$

$$\text{Quick Ratio} = \frac{\text{CE} + \text{MS} + \text{AR}}{\text{Current Liability}}$$

Where:

CE = Cash Equivalents

MS = Marketable Security

AR = Accounts Receivable

Cash Ratio

- The cash ratio is a liquidity measure that shows a company's ability to cover its short-term obligations using only cash and cash equivalents.
- The cash ratio is derived by adding a company's total reserves of cash and near-cash securities and dividing that sum by its total current liabilities.
- The cash ratio is more conservative than other liquidity ratios because it only considers a company's most liquid resources.
- A calculation greater than 1 means a company has more cash on hand than current debts, while a calculation less than 1 means a company has more short-term debt than cash.
- Lenders, creditors, and investors use the cash ratio to evaluate the short-term risk of a company.

$$\text{Cash Ratio} = \frac{\text{CE}}{\text{Current Liability}}$$

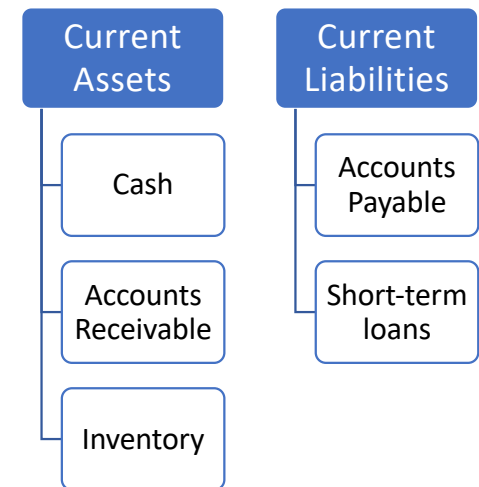
Where:

CE = Cash & Cash Equivalents

Working Capital

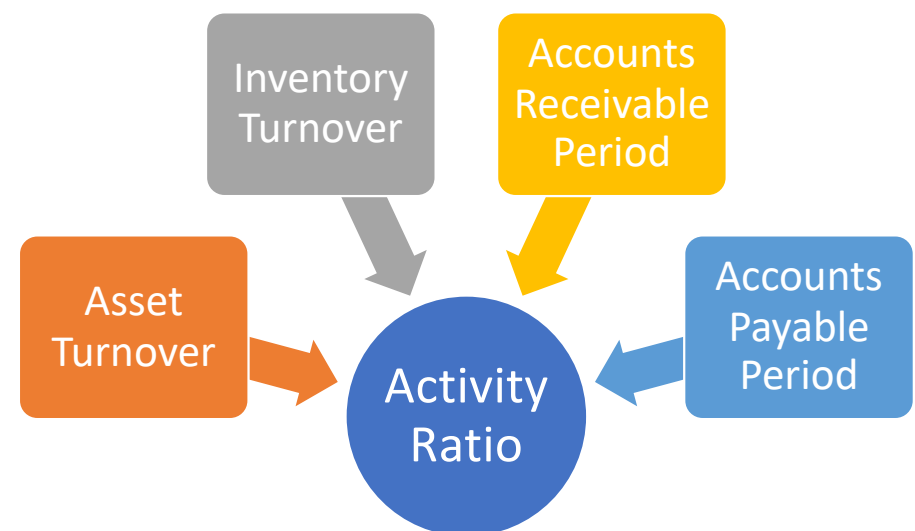
- Working capital is a measure of a company's liquidity, operational efficiency, and short-term financial health.
- A company has negative working capital if its ratio of current assets to liabilities is less than one.
- Positive working capital indicates that a company can fund its current operations and invest in future activities and growth.
- High working capital isn't always a good thing. It might indicate that the business has too much inventory or is not investing its excess cash.

Working Capital = Current Assets – Current Liabilities



Efficiency or Activity Ratio

- An activity ratio broadly describes any type of financial metric that helps investors gauge how efficiently a company uses its assets to generate revenues and cash.
- Activity ratios may be utilized to compare two different businesses within the same sector, or they may be used to monitor a single company's fiscal health over time.
- Activity ratios can be subdivided into merchandise inventory turnover ratios, total assets turnover ratios, return on equity measurements, and a spectrum of other metrics.



Asset Turnover

- Asset turnover is the ratio of total sales or revenue to average assets.
- This metric helps understand how effectively companies are using their assets to generate sales.
- Investors use the asset turnover ratio to compare similar companies in the same sector or group.
- A company's asset turnover ratio can be impacted by large asset sales as well as significant asset purchases in a given year.

$$\text{Asset Turnover} = \frac{\text{Total Sales}}{\text{Average Assets}}$$

$$\text{Average Assets} = \frac{\text{Asset at Start of year} + \text{Asset at End of year}}{2}$$

Inventory Turnover

- Inventory includes all the goods a company has in its stock that will ultimately be sold.
- Inventory turnover indicates the rate at which a company sells and replaces its stock of goods during a particular period.
- The higher the inventory turnover, the better, since high inventory turnover typically means a company is selling goods quickly, and there is considerable demand for their products.
- Inventory turnover indicates how well a company is managing its stock.
- Inventory turnover also shows whether a company's sales and purchasing departments are in sync.
- Days sales of inventory (DSI) measures how many days it takes for inventory to turn into sales. For example if DSI is 90 then a company takes this much days to sell his product.

$$\text{Inventory Turnover Ratio} = \frac{\text{Cost of Goods Sold}}{\text{Average Inventory}}$$

$$\text{Average Inventory} = \frac{\text{Inv. at Start of year} + \text{Inv. at End of year}}{2}$$

$$\text{Days sales Inventory} = \frac{\text{Average Inventory}}{\text{Cost of Goods Sold}} \times 365$$

Accounts Receivable Turnover

- The accounts receivable turnover ratio is an accounting measure used to quantify a company's effectiveness in collecting its receivables or money owed by clients.
- A high receivables turnover ratio may indicate that a company's collection of accounts receivable is efficient and that the company has a high proportion of quality customers that pay their debts quickly.
- A low receivables turnover ratio could be the result of inefficient collection, inadequate credit policies, or customers who are not financially viable or creditworthy.
- A company's receivables turnover ratio should be monitored and tracked to any trends or patterns.

$$\text{Accounts Receivable Turnover} = \frac{\text{Net Credit Sales}}{\text{Average Accounts Receivable}}$$

$$\text{Average A.R.} = \frac{\text{A.R. at Start of year} + \text{A.R. at End of year}}{2}$$

$$\text{Days collection period} = \frac{\text{Average Accounts Receivable}}{\text{Net Credit Sales}} \times 365$$

Days a client take
to pay the seller.

Accounts Payable Turnover

- The accounts payable turnover ratio is a short-term liquidity measure used to quantify the rate at which a company pays off its suppliers.
- Accounts payable turnover shows how many times a company pays off its accounts payable during a period.
- Ideally, a company wants to generate enough revenue to pay off its accounts payable quickly, but not so quickly the company misses out on opportunities because they could use that money to invest in other endeavours.

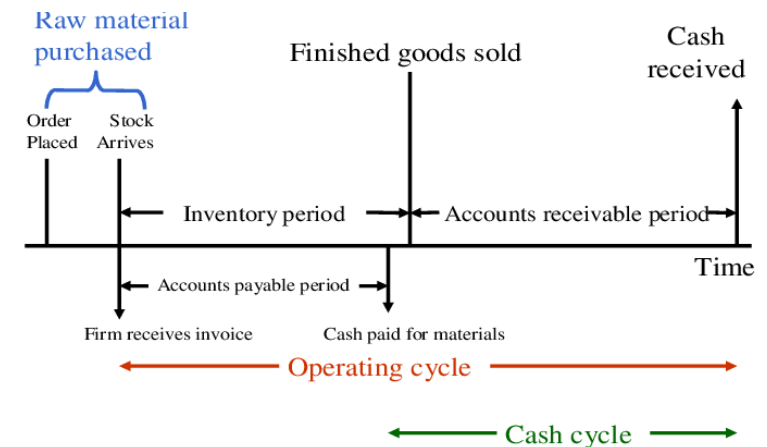
$$\text{Accounts Payable Turnover} = \frac{\text{Total Supply Purchase}}{\text{Average Accounts Payable}}$$

$$\text{Average A.P.} = \frac{\text{A.P. at Start of year} + \text{A.P. at End of year}}{2}$$

$$\text{Days payable period} = \frac{\text{Average Accounts Payable}}{\text{Total Supply Purchase}} \times 365$$

Operating Cycle/Cash Conversion Cycle

- The cash conversion cycle (CCC) is a metric that expresses the length of time (in days) that it takes for a company to convert its investments in inventory and other resources into cash flows from sales.
- This metric takes into account the time needed to sell its inventory, the time required to collect receivables, and the time the company is allowed to pay its bills without incurring any penalties.
- CCC will differ by industry sector based on the nature of business operations.



Operating Cycle =

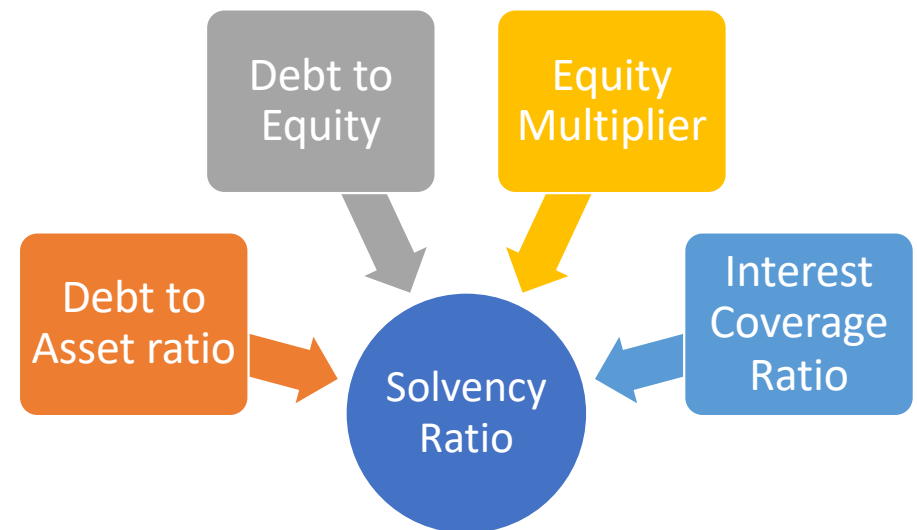
Days Sales Inventory

+ Days Collection Period

- Days Payable Period

Solvency/Leverage Ratio

- A solvency ratio examines a firm's ability to meet its long-term debts and obligations.
- Solvency ratios are often used by prospective lenders when evaluating a company's creditworthiness as well as by potential bond investors.
- Solvency ratios and liquidity ratios both measure a company's financial health but solvency ratios have a longer-term outlook than liquidity ratios.



Debt to Asset Ratio/Debt Ratio

- The debt-to-assets ratio shows the degree to which a company has used debt to finance its assets.
- The calculation considers all of the company's debt, not just loans and bonds payable, and considers all assets, including intangibles.
- If a company has a debt-to-assets ratio of 0.4, 40% of its assets are financed by creditors, and 60% are financed by owners (shareholders) equity.

$$\text{Debt/Asset Ratio} = \frac{\text{Short-term debt} + \text{Long-term debt}}{\text{Total Assets}}$$

Debt to Equity Ratio

- The D/E ratio is an important metric used in corporate finance.
- It is a measure of the degree to which a company is financing its operations through debt versus wholly owned funds.
- More specifically, it reflects the ability of shareholder equity to cover all outstanding debts in the event of a business downturn.
- It can be used to evaluate how much leverage(debt) a company is using.
- Higher-leverage ratios tend to indicate a company or stock with higher risk to shareholders.

$$\text{Debt/Equity Ratio} = \frac{\text{Total Liabilities}}{\text{Shareholder Equity}}$$

D/E ratio is an important measure of capital structure.

Equity Multiplier

- The equity multiplier is a risk indicator that measures the portion of a company's assets that is financed by stockholder's equity rather than by debt.
- Generally, a high equity multiplier indicates that a company is using a high amount of debt to finance assets. A low equity multiplier means that the company has less reliance on debt.
- A company's equity multiplier can be seen as high or low only in comparison to historical standards, the averages for the industry, or the company's peers.

$$\text{Equity Multiplier} = \frac{\text{Total Assets}}{\text{Shareholder Equity}}$$

Interest Coverage Ratio

- The interest coverage ratio is a debt and profitability ratio used to determine how easily a company can pay interest on its outstanding debt.
- This ratio is used by creditors and prospective lenders to assess the risk of lending capital to a firm.
- A higher coverage ratio is better, although the ideal ratio may vary by industry.
- It measures how many times a company can cover its current interest payment with its available earnings.
- it measures the margin of safety a company has for paying interest on its debt during a given period.

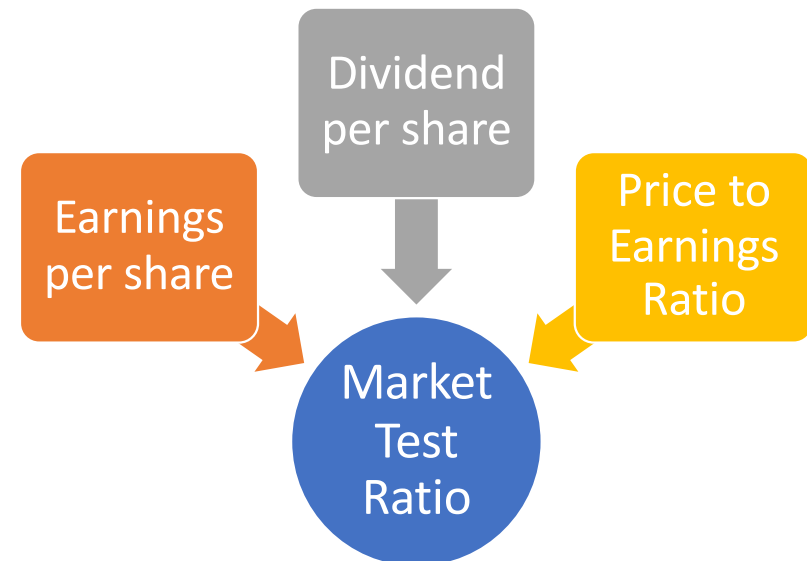
$$\text{Interest Coverage Ratio} = \frac{\text{EBIT}}{\text{Interest Expense}}$$

Where:

EBIT = Earnings before Interest & Tax

Market Test Ratio

- Market value ratios are used to evaluate the current share price of a publicly-held company's stock.
- These ratios are employed by current and potential investors to determine whether a company's shares are over-priced or under-priced.
- CEO must be able to see the company's performance from the perspective of investors, and so is much more likely to track these measurements closely.
- Market value ratios are not applied to the shares of privately-held entities, since there is no accurate way to assign a market value to their shares.



Earnings per share

- Earnings per share (EPS) is a company's net profit divided by the number of common shares it has outstanding.
- EPS indicates how much money a company makes for each share of its stock and is a widely used metric for estimating corporate value.
- A higher EPS indicates greater value because investors will pay more for a company's shares if they think the company has higher profits relative to its share price.

$$\text{Earnings Per Share} = \frac{\text{Net Income}}{\text{Number of outstanding shares}}$$

Book value per share

- Book value per share (BVPS) takes the ratio of a firm's common equity divided by its number of shares outstanding.
- Book value of equity per share effectively indicates a firm's net asset value (total assets - total liabilities) on a per-share basis.
- When a stock is undervalued, it will have a higher book value per share in relation to its current stock price in the market.
- BVPS is used mainly by stock investors to evaluate a company's stock price.

$$\text{Book value Per Share} = \frac{\text{Equity of common shareholder}}{\text{Number of outstanding shares}}$$

Price to Earnings Ratio

- The price-to-earnings ratio (P/E ratio) is the ratio for valuing a company that measures its current share price relative to its per-share earnings (EPS)
- The price-earnings (P/E) ratio relates a company's share price to its earnings per share.
- A high P/E ratio could mean that a company's stock is over-valued, or else that investors are expecting high growth rates in the future.
- Companies that have no earnings or that are losing money do not have a P/E ratio since there is nothing to put in the denominator.

$$\text{P/E Ratio} = \frac{\text{Market value per share}}{\text{Earning per share(EPS)}}$$

Dividend per share

- Dividend per share (DPS) is the sum of declared dividends issued by a company for every ordinary share outstanding.
- DPS is calculated by dividing the total dividends paid out by a business, including interim dividends, over a period of time, usually a year, by the number of outstanding ordinary shares issued.
- DPS is an important metric to investors because the amount a firm pays out in dividends directly translates to income for the shareholder.
- A growing DPS over time can also be a sign that a company's management believes that its earnings growth can be sustained.

$$\text{Dividends Per Share} = \frac{\text{Dividends paid over a period}}{\text{Number of outstanding shares}}$$

Thank You